

Three-Layer Modular Software Architecture for Safety-Critical Human-Robot Collaboration

Integration Layer

RESTful API (OpenAPI 3.0)
WebSocket Streaming (50Hz)
OPC-UA Server (IEC 62541)
Modbus/TCP Gateway
EtherCAT Master

API Calls

Real-Time Control Layer

Learned Control Barrier Functions (1.8ms inference)
Model Predictive Control (2.8ms QP solve)
Sensor Fusion (4.5 ms, concurrent thread)
Safety Monitor (6.2 ms critical path @ 50 Hz)
Fault Detection & Recovery

Control Cmds

Physical Abstraction Layer

IRobotController (Adapter Pattern)
- URRobotController (RTDE 125Hz)
- KUKARobotController (FRI 200Hz)
- ABBRobotController (RWS REST)
Sensor Array (6× RealSense D435)
Hardware Abstraction

API latency

2.1 ms (p50) / 4.8 ms (p95)

Control critical path

6.2 ms @ 50 Hz

Cross-vendor code reuse

86%

Simulated availability

98.5%